Ever since the British introduced it widely, the invasive water hyacinth has suffocated wetlands across India. The weed erodes biodiversity, ruins aquatic life and, as a corollary, also chokes livelihoods. Assam's Deepor Beel, a wetland near Guwahati, was precisely such a watery graveyard. Now, a bold innovation flips the script: Kumbhi Kagaz, paper crafted from this very ecological curse. Co-founder and wildlife enthusiast Rupankar Bhattacharjee, 28, was assisting his mentor, renowned herpetologist Jayaditya Purkayastha, track a python at Deepor Beel when the snake slipped away into a thicket of water hyacinth, ruining his photo session. As Purkayastha deplored the choked wetland, Rupankar set about thinking. An idea took root when he confided with his friend and co-founder, Aniket Dhar, 25. During the COVID-19 lockdown, Rupankar researched further and set upon a crucial fact: the plant's high fibre content was perfect for paper production. From that eureka moment was born a business model. Kumbhi Kagaz came about in 2022, named after 'jal kumbhi', the Hindi name for water hyacinth.

Securing €10,000 (around Rs 8.3 lakhs) from the Zero Waste City Challenge by WasteAid, UK, Rupankar and Aniket established their factory in Guwahati's Azara. The enterprise achieved an annual turnover of Rs 15.5 lakh in the past financial year. Rupankar oversees production and machinery, while Aniket manages business operations, supported by Purkayastha's wife, Madhurima Das, who handles the research and scientific aspects.

What sets Kumbhi Kagaz apart is its pioneering approach. While other attempts at making paper from water hyacinth blended it with other materials, Kumbhi Kagaz perfected a unique process to create "100 per cent" water hyacinth paper. "We don't use any chemicals. We don't use any acid," says Aniket Dhar. The secret is their custom-designed mixture machine, which efficiently breaks down fibres, allowing them to bind cohesively.

The venture has breathed life back into the Deepor Beel ecosystem, restoring livelihoods along with the environment. Since 2023, it has removed 42 tonnes of water hyacinth from Deepor Beel. Its efforts have now reached the wetlands of Kaziranga National Park and Goalpara's Urpad Beel—saving local fishing communities from costly cleanup operations.

The company struggled in its early days. With no background in production, the founders had to double up as artisans and mechanics, learning the craft on the go. Humidity in the Northeast posed another hurdle, making the drying process painfully slow and restricting output. Support arrived just in time: an investment from Assam Startup, a central government grant under the Amrut 2.0 competition of the Union ministry of housing and urban affairs, and additional funding of Rs 54 lakh from the GAP Fund, backed by the International Fund for Agricultural Development (IFAD). National recognition came when PM Narendra Modi mentioned Kumbhi Kagaz on his *Mann Ki Baat* programme. Offers to expand across India followed, but the founders have chosen to first anchor themselves in Assam. Their vision is to secure a Geographical Indication (GI) tag for water hyacinth handicrafts, making it an emblem of Assam, alongside tea.

The roadmap is clear: the first two years will focus on building a strong presence in the stationery market, making ecofriendly paper affordable. Beyond stationery, the next frontier will be packaging. "Right now, our paper costs a little more than the regular product. We want to level that to make sustainability accessible. Eco-friendly purchases shouldn't be a luxury," says Aniket. ■

> **"OUR PAPER COSTS A BIT MORE THAN REGULAR ONES. WE WANT TO LEVEL THAT SO AS TO MAKE IT ACCESSIBLE, AS ECO-FRIENDLY BUYS SHOULDN'T BE A LUXURY"**
>
> **ANIKET DHAR**

FACT SHEET

Name of company
Kumbhi Kagaz

What they do
Make fully eco-friendly paper from water hyacinth

Co-founders
Rupankar Bhattacharjee, 28
Madhurima Das, 45

Founded in: 2022

Annual turnover
Rs 15.5 lakh